

Chapter 1 – Introduction to Windows 11

Learning Objectives

After completing this chapter, you will be able to:

- Understand what Windows 11 is.
 - Learn why Windows 11 is widely used.
 - Identify the different Windows 11 editions.
 - Understand the Windows architecture.
 - Recognize Windows' role in cybersecurity and Blue Team operations.
 - Understand why Windows is commonly monitored by Security Operations Centers (SOCs).
 - Prepare for building a Windows 11 virtual machine in the following chapters.
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1.1 What is Windows 11?

Windows 11 is Microsoft's latest desktop operating system, designed to run on personal computers, laptops, and virtual machines. It provides a graphical interface that allows users to interact with hardware, run applications, manage files, connect to networks, and perform everyday computing tasks.

As an operating system, Windows 11 acts as the bridge between the user, software applications, and the computer's hardware.

It is widely used in:

- Homes
- Schools
- Universities
- Businesses
- Government organizations

- Healthcare
- Financial institutions

Because of its widespread adoption, Windows is one of the most targeted operating systems for cyberattacks. This also makes it one of the most important platforms for cybersecurity professionals to understand.

1.2 Why Learn Windows 11?

Many enterprise environments use Windows devices as employee workstations.

As a cybersecurity professional, especially in Blue Team roles, you will frequently investigate:

- Failed login attempts
- Malware infections
- PowerShell activity
- User account changes
- File modifications
- Windows Event Logs
- Windows Defender alerts

Understanding Windows is essential for:

- SOC Analysts
 - Incident Responders
 - System Administrators
 - Digital Forensics Analysts
 - Threat Hunters
 - Security Engineers
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1.3 Why Use Windows 11 in a Blue Team Lab?

A Blue Team lab should simulate a real organizational environment.

Most organizations have:

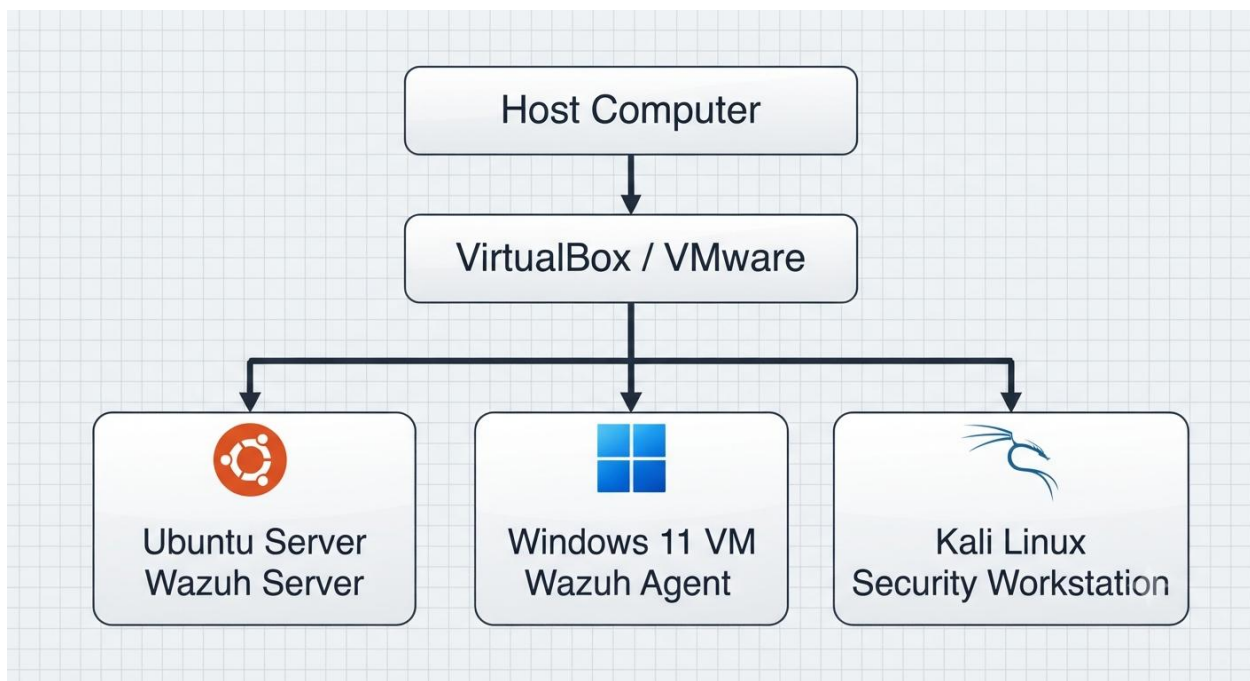
- Windows endpoints
- Linux servers
- Network devices
- Security monitoring tools

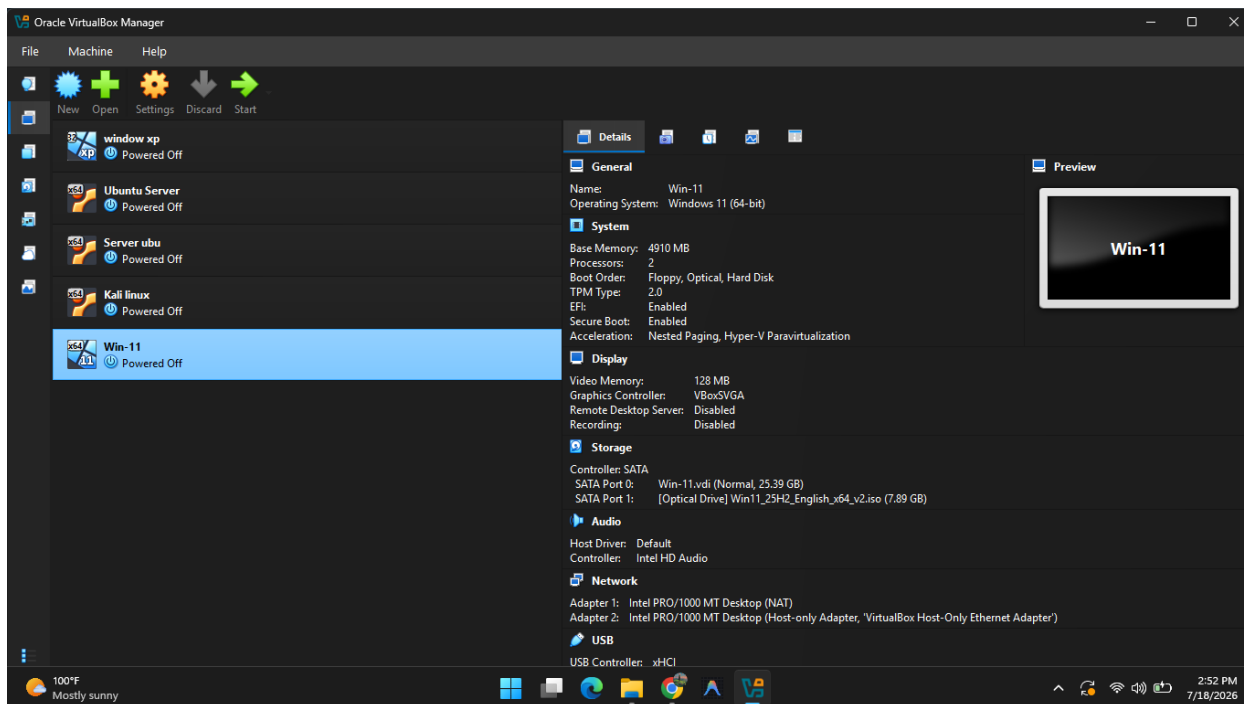
In our home lab:

- **Ubuntu Server** hosts the Wazuh platform.
- **Windows 11** acts as the monitored endpoint.
- **Kali Linux** is used for security testing and administration.

This allows us to observe how Windows activity is collected, analyzed, and displayed by Wazuh.

Home Lab Architecture





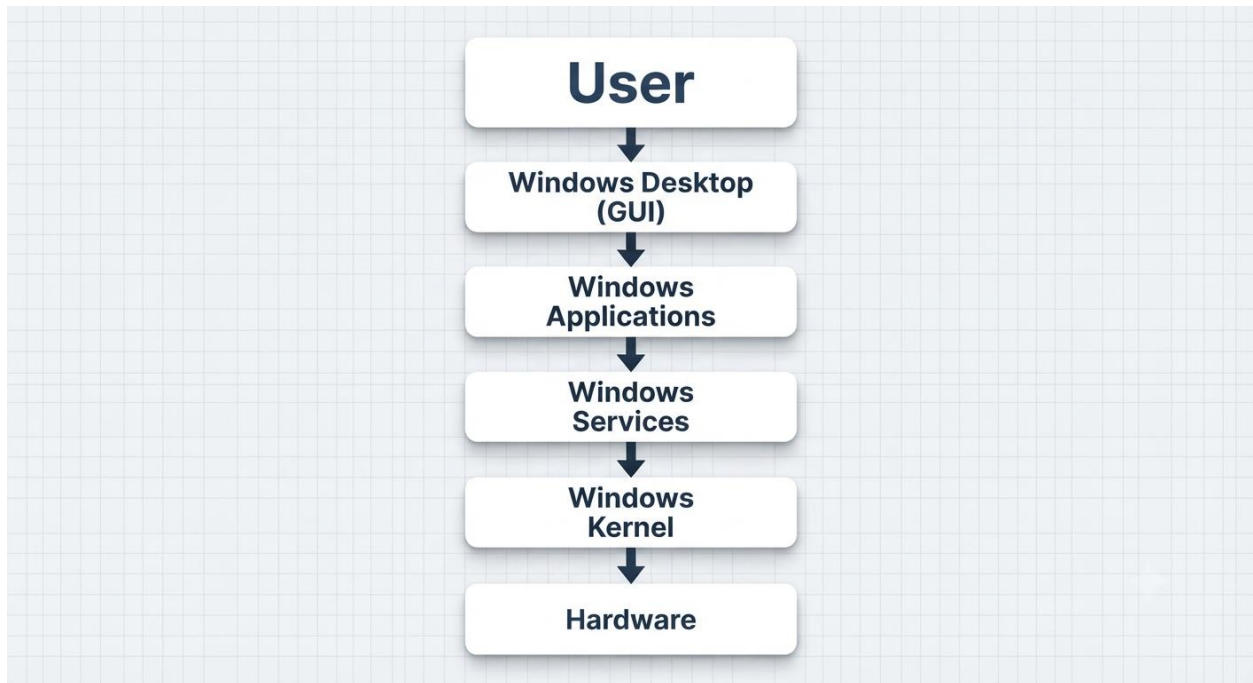
1.4 Windows 11 Editions

Microsoft provides several editions of Windows 11.

Edition	Intended Use
Home	Personal computers
Pro	Professionals and small businesses
Enterprise	Large organizations
Education	Schools and universities

For this handbook, we recommend using **Windows 11 Pro**, as it includes features commonly used in enterprise environments, such as BitLocker, Group Policy, and Remote Desktop.

1.5 Windows Architecture



Each layer has a specific responsibility:

- **GUI** – Allows users to interact with the system.
- **Applications** – Software such as browsers, Office, and development tools.
- **Services** – Background processes that support Windows functionality.
- **Kernel** – The core of the operating system that manages hardware resources.
- **Hardware** – CPU, memory, storage, and other physical components.

1.6 Key Features of Windows 11

Windows 11 introduces several improvements over previous versions.

Feature	Purpose
Modern User Interface	Simplified and cleaner desktop experience
Windows Security	Built-in antivirus and security management
Microsoft Defender	Protects against malware and threats
BitLocker*	Encrypts storage drives (<i>Pro edition</i>)
Virtual Desktops	Organize workspaces
Windows Update	Keeps the operating system current
Event Viewer	Displays system and security events



1.7 Windows Security Features

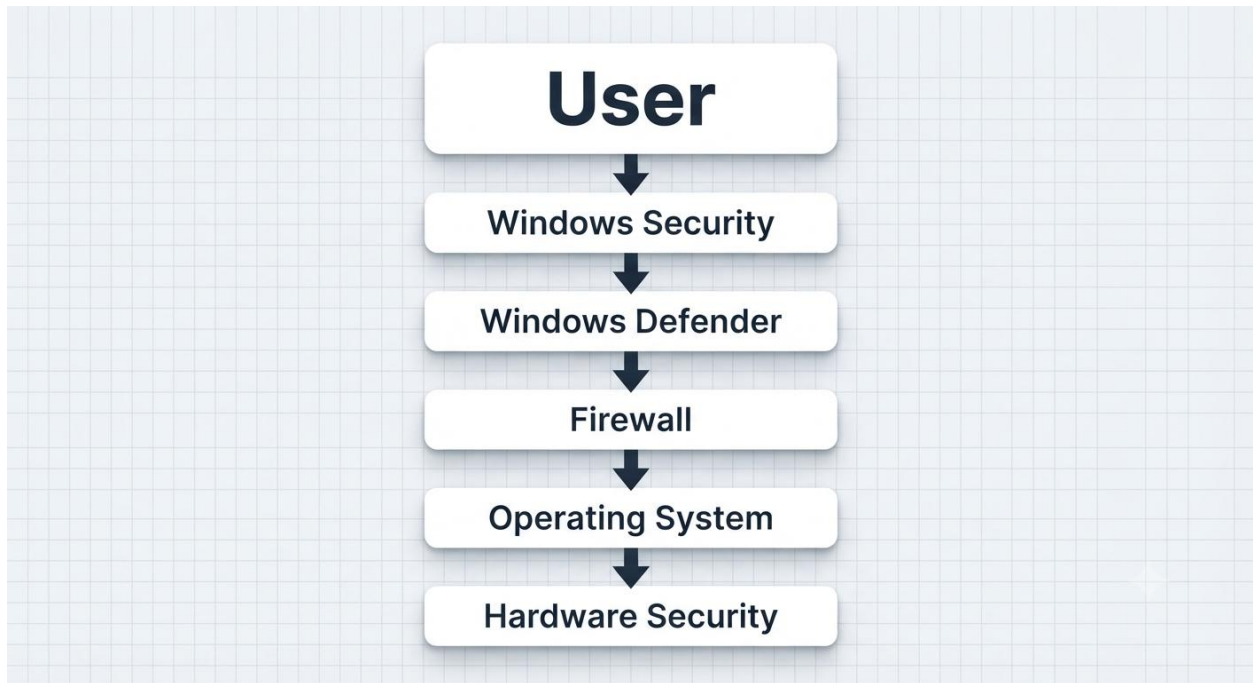
Windows includes several built-in security technologies.

Some important features are:

- Microsoft Defender Antivirus
- Windows Firewall
- User Account Control (UAC)
- SmartScreen
- BitLocker (Pro edition)
- Secure Boot
- TPM (Trusted Platform Module)

These features help protect systems from malware, unauthorized access, and other security threats.

Security Layers



1.8 Windows in Security Operations

A Security Operations Center (SOC) continuously monitors endpoints for suspicious activity.

Windows systems generate a large amount of security-relevant information, including:

- Login events
- User account changes
- Software installations
- Service status changes
- PowerShell activity

- Windows Defender alerts
- Event Logs

These events help analysts identify and investigate potential threats.

Security Monitoring Workflow



1.9 Windows Event Logs

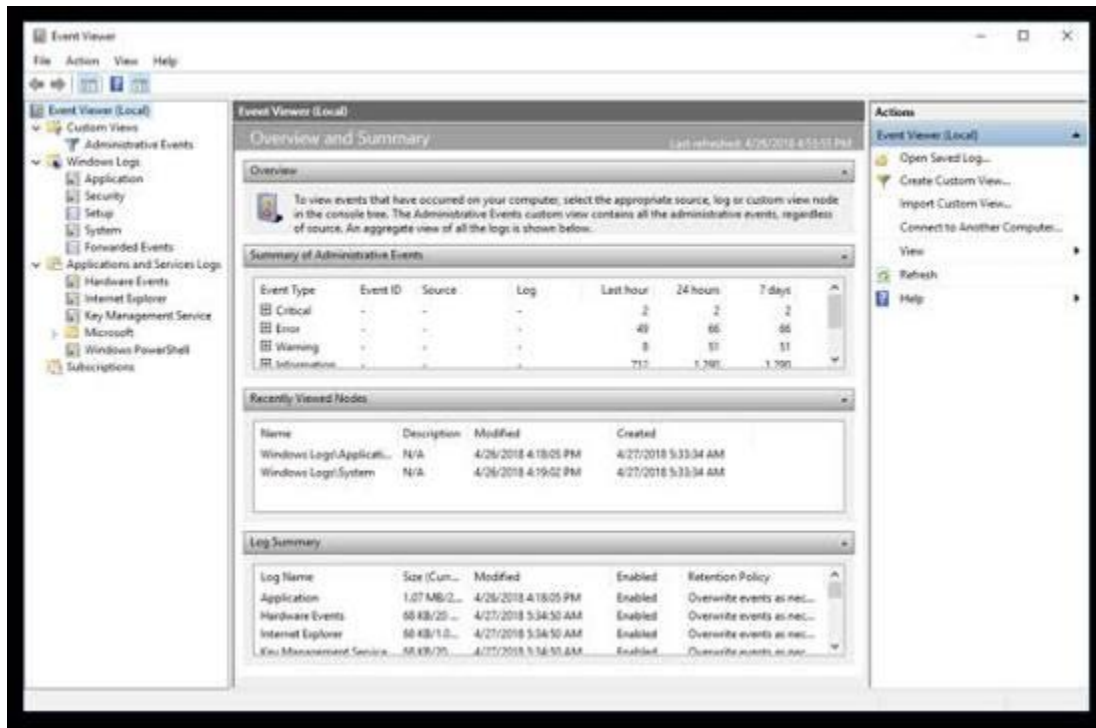
One of the most valuable sources of information in Windows is the Event Log.

Windows records events related to:

- User authentication
- System startup and shutdown
- Application activity
- Driver issues
- Security events

- Errors and warnings

Later in this handbook, we will learn how Wazuh collects and analyzes these logs.



Event Log Categories

Category	Description
Application	Events from installed applications
Security	Authentication and security-related events
System	Operating system events
Setup	Installation and update events
Forwarded Events	Events collected from remote systems

1.10 Why Cybersecurity Professionals Learn Windows

Most cyber incidents investigated by SOC analysts involve Windows systems.

Examples include:

- Brute-force login attempts
- Malware infections

- Suspicious PowerShell execution
- Unauthorized account creation
- Privilege escalation
- Persistence mechanisms
- Ransomware activity

Knowing how Windows works makes it easier to identify abnormal behavior.

1.11 Ethical and Legal Considerations

Use Windows only in environments where you have authorization to perform security monitoring and testing.

For this handbook:

- Use your personal lab.
- Use your own virtual machines.
- Do not monitor or modify systems you do not own or have permission to access.

Responsible use of security tools is a fundamental part of cybersecurity.

Best Practices

- Use **Windows 11 Pro** for your lab environment.
 - Keep Windows updated with the latest security patches.
 - Use strong passwords for local accounts.
 - Take a virtual machine snapshot after installation.
 - Avoid installing unnecessary software in the lab.
 - Document any configuration changes you make.
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Chapter Summary

In this chapter, you learned what Windows 11 is, why it is important in modern organizations, and how it fits into a Blue Team home lab. You explored Windows editions, core architecture, built-in security features, and the role of Windows Event Logs in security monitoring. You also saw how Windows integrates with Wazuh to provide visibility into endpoint activity.

This foundation prepares you for creating and configuring a Windows 11 virtual machine in the following chapters.

One-Line Summary

Windows 11 is the primary endpoint operating system in many organizations, making it an essential platform for Blue Team professionals to learn, secure, and monitor as part of a modern cybersecurity environment.
